The empirical recurrence for OEIS A233069, proved by transfer matrix

Adrian Perez Fontelles
Independent researcher

1 September 2026

Abstract

OEIS A233069 counts arrays over $\{0,\ldots,3\}$ under a neighbour condition together with two symmetry-breaking clauses: the upper-left entry is 0, and 1 must appear before 2 in row major order. The last of these is not local — it compares first occurrences across the whole array — but it becomes local once the state carries which of the two values was seen first. With that, $a(n)$ is a walk count on $S = 768$ vertices and the empirical recurrence of order 11 contributed by R. H. Hardin follows from a finite exact computation.

2020 Mathematics Subject Classification. 05A15, 05B45, 15B36.

1 The sequence and the conjecture

OEIS A233069 is "Number of nX4 0..3 arrays with no element x(i,j) adjacent to value 3-x(i,j) horizontally, antidiagonally or vertically, top left element zero, and 1 appearing before 2 in row major order.". It has offset 1 and begins

$$14, 289, 6854, 164495, 3957778, 95264272, 2293174089, 55201144642, \ldots$$

The entry carries the following comment:

> Empirical: a(n) = 30*a(n-1) -155*a(n-2) +312*a(n-3) -393*a(n-4) +247*a(n-5) +43*a(n-6) -112*a(n-7) +15*a(n-8) +22*a(n-9) -9*a(n-10) +a(n-11) for n>12

As of the "Last modified" line on the live entry (Jul 23 2025 07:50:55, revision 6) the statement is still recorded as empirical, and nothing on the entry records it as settled either way.

Written out, the claim is

$$a(n) \;=\; 30\,a(n-1)-155\,a(n-2)+312\,a(n-3)-393\,a(n-4)+247\,a(n-5)+43\,a(n-6)-112\,a(n-7)+15\,a(n-8)+2 \tag{1}$$

for all $n > 12$.

2 The three clauses

The arrays have 4 columns and $L = n$ rows; write $x_{t,u}$ for the entry in row t and column u. The neighbour condition is

$$x_{t+d_1,\,u+d_2} \neq 3 - x_{t,u} \qquad \text{for every offset } (d_1,d_2) \in \mathcal{N} = \{(0,1),\ (1,-1),\ (1,0)\} \tag{2}$$

inside the array. The relation "$y = 3 - x$" is symmetric, so each unordered adjacency need only be tested once; the offsets above are the forward representatives, one per direction, and every one of them has a nonnegative row shift. A single row of state is therefore enough for (2): the cells of a row are settled by that row and the next.

Here the walk runs along rows, which is the order row major reading uses, so a single flag suffices: 0 while neither 1 nor 2 has occurred, and thereafter 1 or 2 according to which occurred first, the first occurrence inside a row being found by scanning it left to right. An array is accepted when the flag is not 2.

Lemma 1. *Let $\mathcal{V}$ be the set of pairs (a row, a flag vector), $|\mathcal{V}| = S = 768$, let $(r, f) \to (s, f')$ be an edge when (2) holds for the cells of r with s beneath it and f' is f updated by s, let $\mathbf{v}$ mark the vertices whose row begins with 0 and whose flag is the one that row alone produces, and $\mathbf{u}$ those whose row satisfies (2) with nothing beneath it and whose flag is accepted. Then*

$$a(n) \;=\; \mathbf{v}^\top N^{\,L-1}\mathbf{u}, \qquad L = n.$$

Proof. A walk of length $L-1$ is a sequence of L rows, and its flag component is determined by them, so walks correspond to arrays. Along the walk the cells of row t are tested once, at the step leaving it, with the correct neighbour row; the cells of the last row are tested by $\mathbf{u}$ with no row beneath, which is how the array behaves at its edge. The flag component is by construction the record of first occurrences among the rows laid down so far, so $\mathbf{u}$ imposes exactly the ordering clause, and $\mathbf{v}$ imposes the upper-left clause. The state carries a row, not a boundary, so L rows make a walk of length $L-1$. ☐

3 The criterion, and the computation

Let $q(t) = t^{11} - \sum_i c_i\, t^{11-i}$ be the characteristic polynomial of (1) and $G = N$.

Lemma 2. *With n_0 the least index for which $L \geq 1$, $o = 1n_0 + 0 - 1$ and $h_j = G^j N^o \mathbf{u}$, we have for $j \geq 11$*

$$a(n_0 + j) - \sum_i c_i\, a(n_0 + j - i) = \mathbf{v}^\top G^{\,j-11} w, \qquad w = h_{11} - \sum_i c_i h_{11-i}.$$

Hence (1) holds from that point on if and only if $\mathbf{v}^\top G^m w = 0$ for all $m \geq 0$, and $m < S$ suffices.

Proof. Substitute Lemma 1 and factor. By Cayley–Hamilton G^S is an integer combination of $I, G, \ldots, G^{S-1}$, so every later value repeats that combination of earlier ones. ☐

Theorem 3. *The recurrence (1) holds for every $n > 12$, which is the statement of Section 1.*

Proof. The digraph was built directly from (2) and the flag update. The vector w was formed by 11 applications of G in exact integer arithmetic and $\mathbf{v}^\top G^m w$ evaluated for $m = 0, 1, \ldots,$ again exactly. Every one of those integers vanishes from the index corresponding to $n = 12 + 1$ onwards, and the run continued until $S + 1$ consecutive zeros had been seen. ☐

4 Verification

Three checks, each independent of the algebra above.

First, the walk counts of Lemma 1 were compared against the entry's DATA at every one of the 15 published indices, with no shift fitted: the number of rows is $L = n$ from the entry's own name and the offset says which index carries the first published term. The values agree exactly. The ordering clause is where a misreading would show first, and it is the reason the flags are per column rather than a single value.

Second, the recurrence (1) was evaluated on those published terms in exact integer arithmetic, with no matrices involved, and vanishes wherever it applies.

Third, the annihilation test of Lemma 2 was carried to the full Cayley–Hamilton bound in exact integer arithmetic rather than to a sampled prefix.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A233069.

[2] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)

[3] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009.

[4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.